

- 1 Scientists are investigating the variation in air pressure at different locations on a mountain.

- (a) The scientists take measurements of several physical quantities at each location.

Complete Table 1.1 by stating the SI base unit for each quantity and identifying with a tick (✓) whether each quantity is a scalar or a vector. Use the space for any working.

Table 1.1

quantity measured	SI base unit	scalar	vector
air temperature			
air pressure			

[2]

- (b) (i) At one location, the density of the air is 1.1 kg m^{-3} . A spherical weather balloon is filled with a gas and released from rest. The balloon has radius 0.90 m.

Calculate the upthrust acting on the balloon when it is released.

upthrust = N [2]

- (ii) Explain why an upthrust acts on the balloon.

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 [2]

